

Physics of Semiconductor Devices

ECE210: PSD



INDRAPRASTHA INSTITUTE *of*
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(week12)



What are the basic transport mechanisms
in a semiconductor?



Drude model (a classical approach of classical conductivity)

The basic assumptions of the Drude model

1. between collisions the interaction of a given electron

independent electron approximation
free electron approximation

2. collisions are instantaneous events

Drude considered electron scattering off
the impenetrable ion cores

the specific mechanism of the electron scattering is not considered below

3. an electron experiences a collision with a probability per unit time $1/\tau$

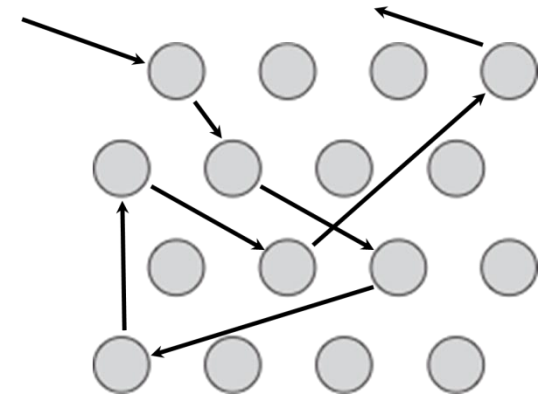
dt/τ – probability to undergo a collision within small time dt

randomly picked electron travels for a time τ before the next collision

τ is known as the relaxation time, the collision time, or the mean free time

τ is independent of an electron position and velocity

4. after each collision an electron emerges with a velocity that is randomly directed and with a speed appropriate to the local temperature



Drude theory: electrical conductivity



If we apply an electric field, the equation of motion is

$$m_e \frac{d\mathbf{v}}{dt} = -e\mathcal{E}$$

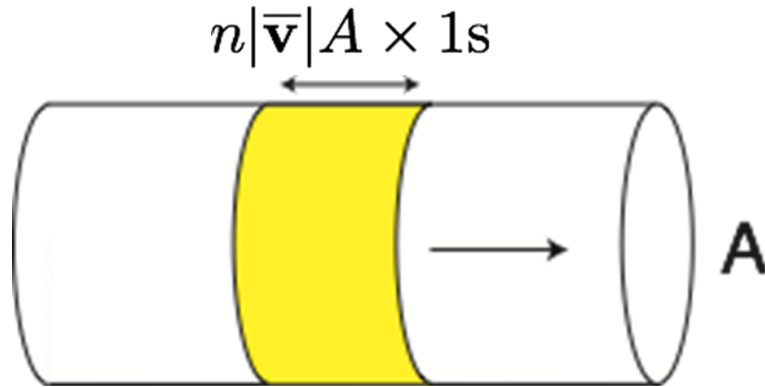
integration gives

$$\mathbf{v}(t) = \frac{-e\mathcal{E}t}{m_e}$$

and if τ is the average time between collisions then the average drift speed is

$$\bar{\mathbf{v}} = \frac{-e\mathcal{E}\tau}{m_e}$$

Drude theory: electrical conductivity



number of electrons passing in unit time

$$n|\bar{\mathbf{v}}|A$$

current of negatively charged electrons

$$-en|\bar{\mathbf{v}}|A$$

current density

$$\mathbf{j} = n\bar{\mathbf{v}}(-e)$$

and with $\bar{\mathbf{v}} = \frac{-e\boldsymbol{\mathcal{E}}\tau}{m_e}$ we get

Ohm's law

$$\mathbf{j} = \frac{ne^2\tau}{m_e}\boldsymbol{\mathcal{E}}$$

Drude theory: electrical conductivity



$$\mathbf{j} = \sigma \mathcal{E} = \frac{\mathcal{E}}{\rho}$$

and we can define, conductivity $\sigma = \frac{ne^2\tau}{m_e} = n\mu e$

Ohm's law

$$\mathbf{j} = \frac{ne^2\tau}{m_e} \mathcal{E}$$

resistivity

$$\rho = \frac{m_e}{ne^2\tau}$$

mobility

$$\mu = \frac{e\tau}{m_e}$$

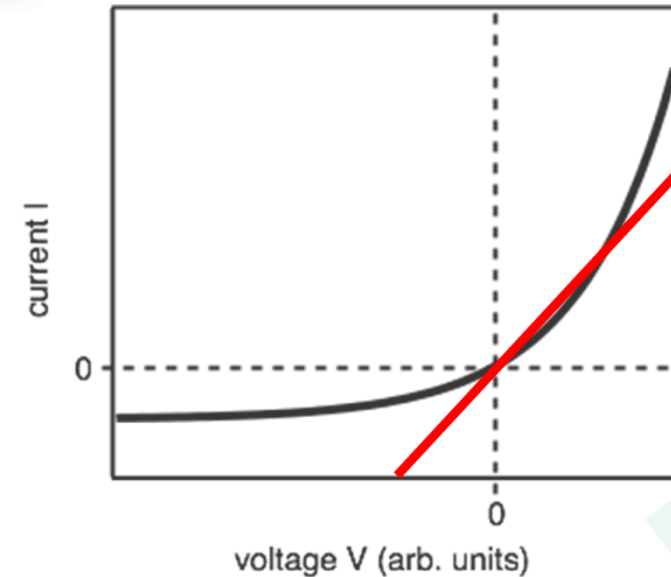
$$|\mathbf{v}| = \mu |\mathcal{E}|$$

Ohm's law

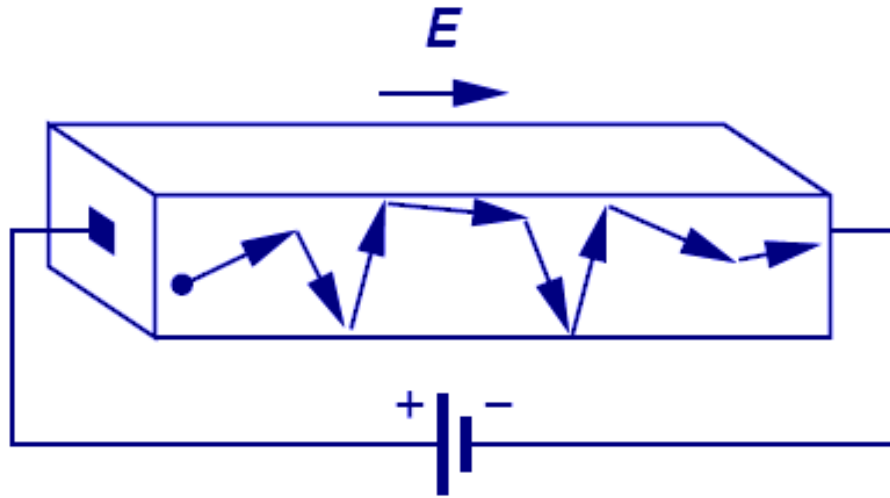


$$\mathbf{j} = \frac{ne^2\tau}{m_e} \boldsymbol{\mathcal{E}}$$

- valid for metals
- valid for homogeneous semiconductors
- not valid for inhomogeneous semiconductors
- not valid for metal contacts to semiconductors



First Charge Transportation Mechanism: Drift



$$\vec{v}_h = \mu_p \vec{E}$$

$$\vec{v}_e = -\mu_n \vec{E}$$

- The process in which charge particles move because of an electric field is called drift.
- Charge particles will move at a velocity that is proportional to the electric field.

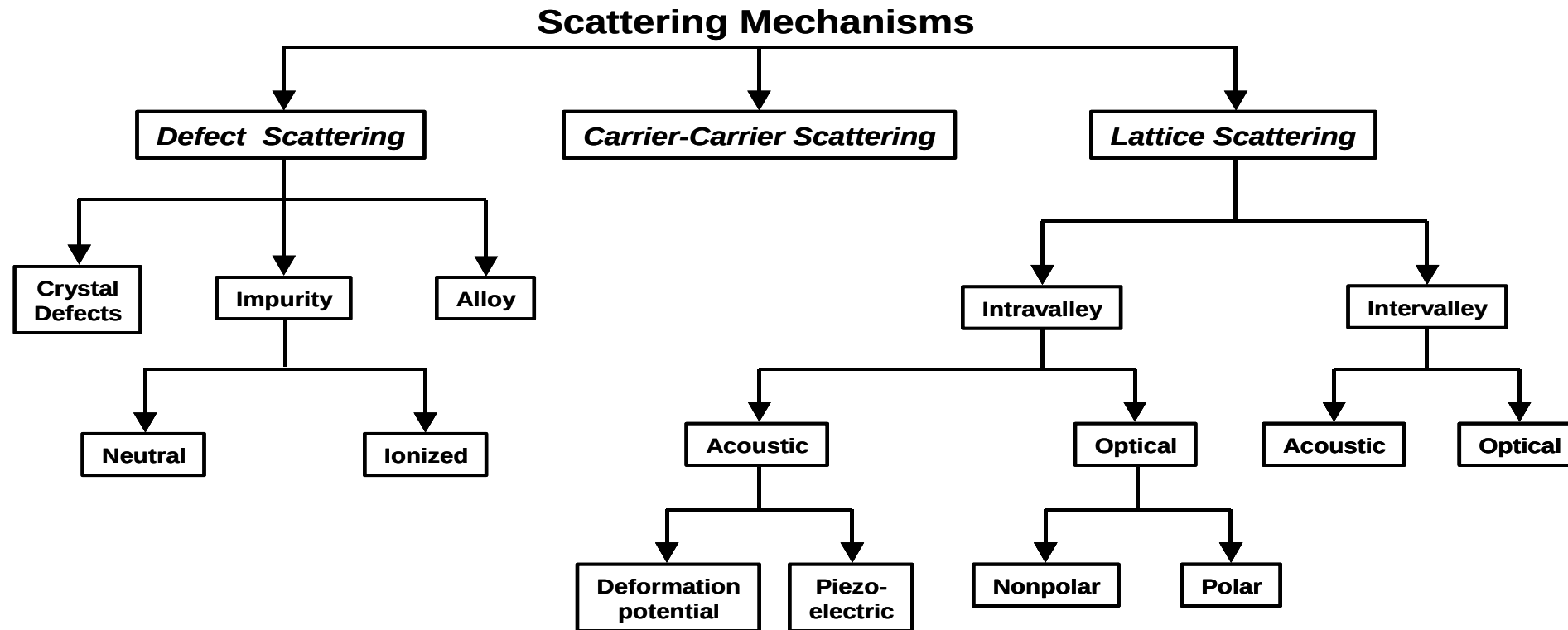
Current Flow: Drift



$$\begin{aligned}J_n &= \mu_n E \cdot n \cdot q \\J_{tot} &= \mu_n E \cdot n \cdot q + \mu_p E \cdot p \cdot q \\&= q(\mu_n n + \mu_p p)E\end{aligned}$$

- Since velocity is equal to μE , drift characteristic is obtained by substituting V with μE in the general current equation.
- The total current density consists of both electrons and holes.

Different scattering mechanism in a semiconductor device



Matthiessen's Rule

$$\frac{1}{\tau} = \frac{1}{\tau_{\text{phonon}}} + \frac{1}{\tau_{\text{impurity}}} \Rightarrow \frac{1}{\mu} = \frac{1}{\mu_{\text{phonon}}} + \frac{1}{\mu_{\text{impurity}}}$$

Mobility

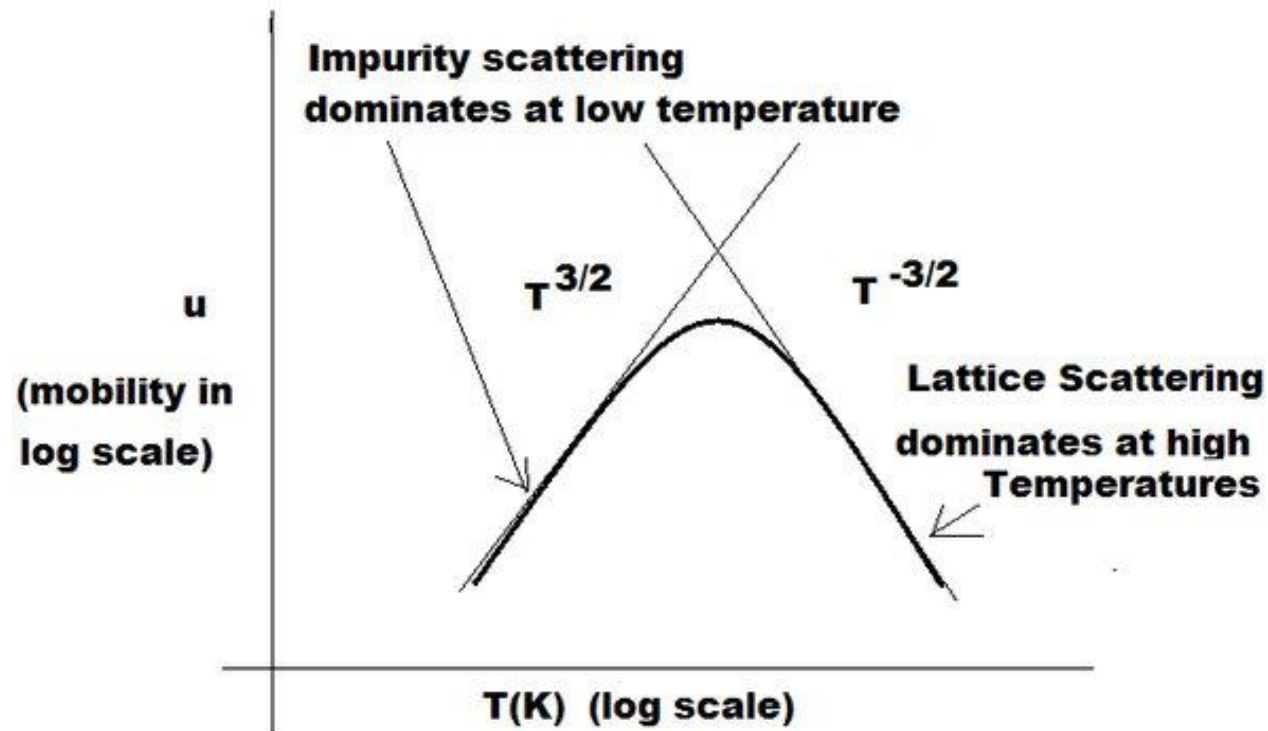
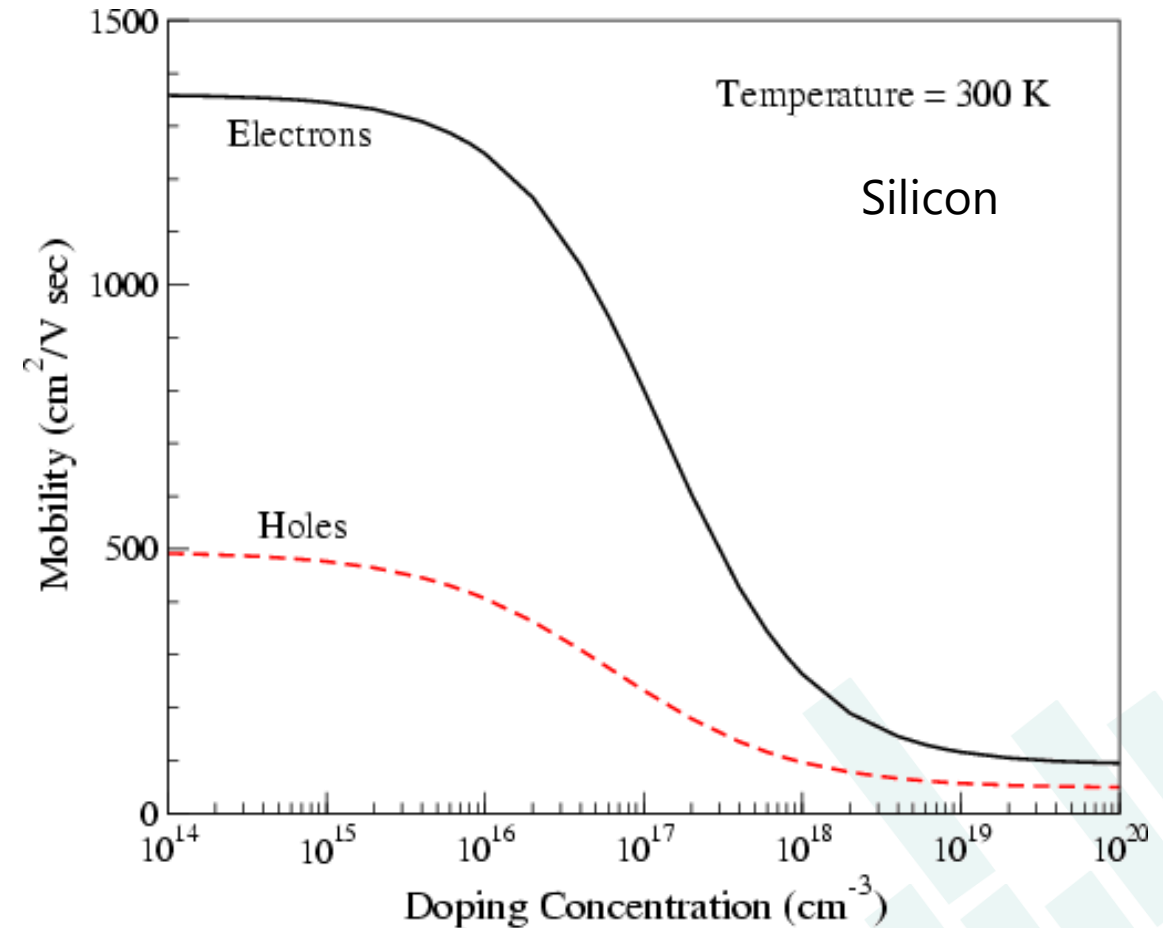
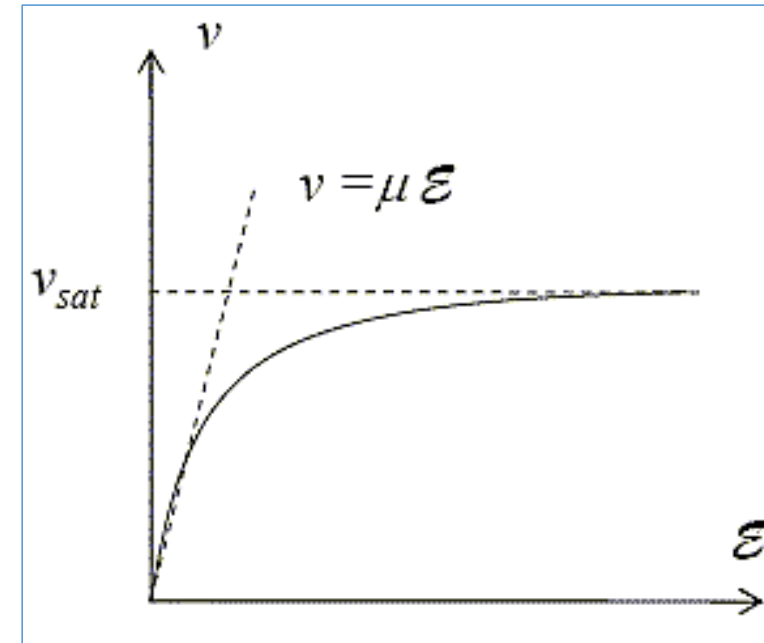
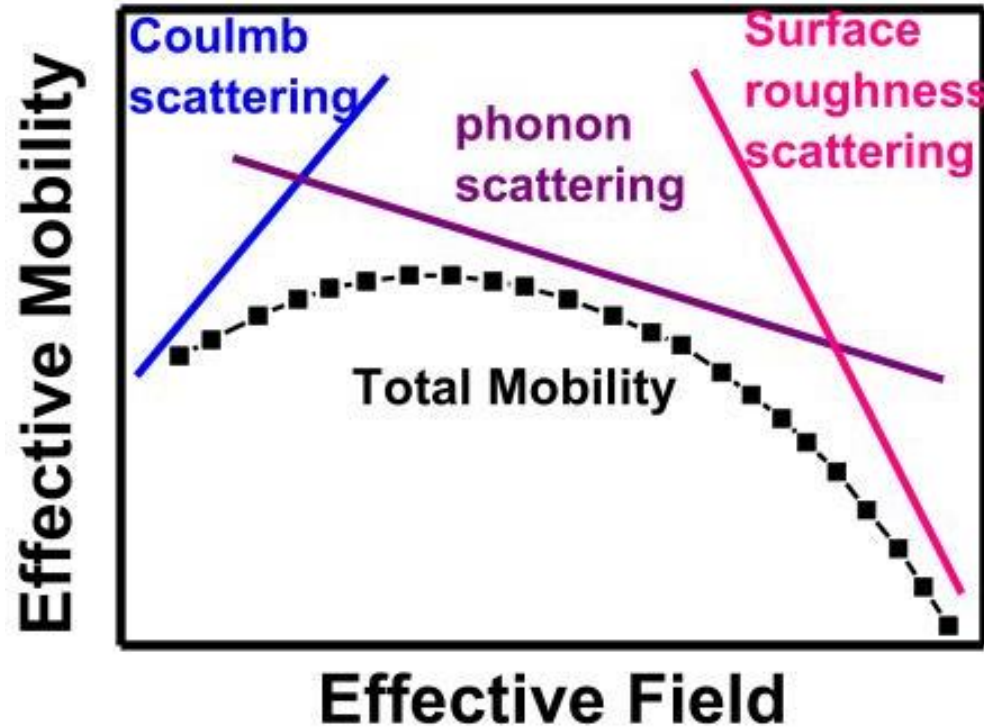


Figure 2.2.38. Temperature dependence of mobility in Si.



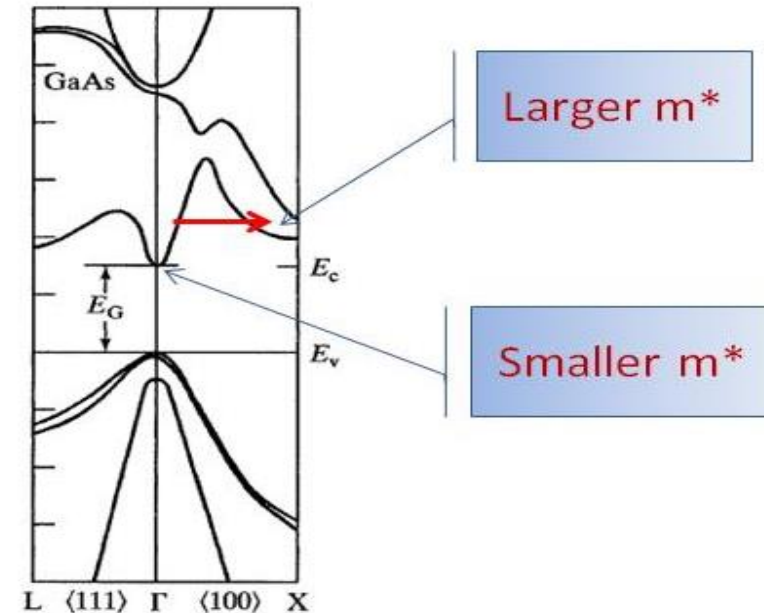
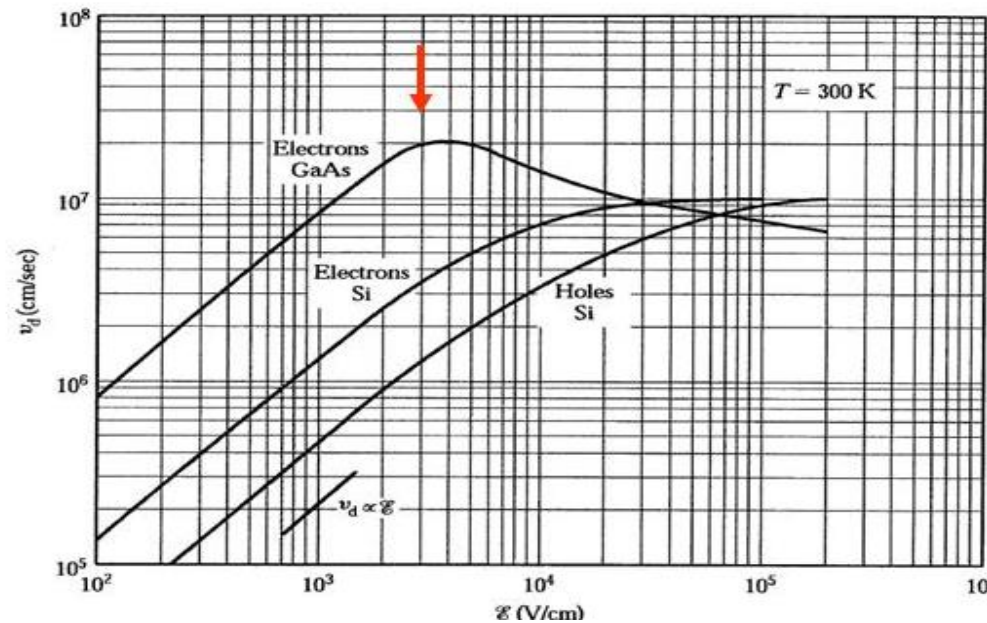
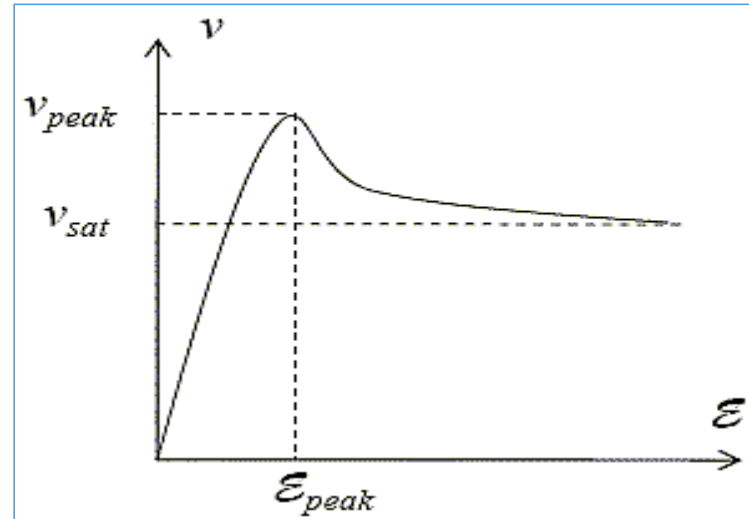
Velocity Saturation



$$v = \frac{\mu_0}{1 + \frac{\mu_0 E}{v_{sat}}} E$$

In reality, velocity does not increase linearly with electric field. It will eventually saturate to a critical value.

Velocity overshoot and inter-valley transfer



Current Flow: Drift



$$\begin{aligned}J_n &= \mu_n E \cdot n \cdot q \\J_{tot} &= \mu_n E \cdot n \cdot q + \mu_p E \cdot p \cdot q \\&= q(\mu_n n + \mu_p p)E\end{aligned}$$

- Since velocity is equal to μE , drift characteristic is obtained by substituting V with μE in the general current equation.
- The total current density consists of both electrons and holes.

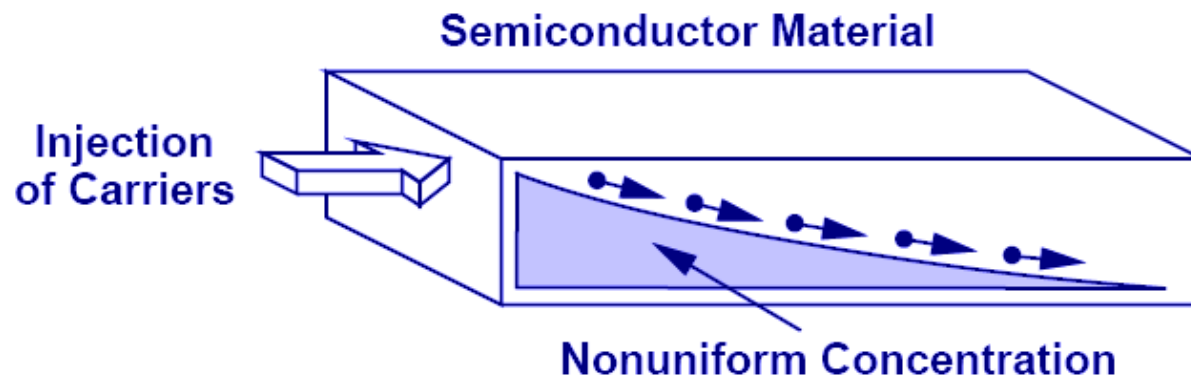
Second Charge Transportation Mechanism: Diffusion



- Due to thermally induced random motion, mobile particles tend to move from a region of high concentration to a region of low concentration.
 - Analogy: ink droplet in water
- Current flow due to mobile charge diffusion is proportional to the carrier concentration gradient.
 - The proportionality constant is the **diffusion constant**.

$$J_n = qD_n \frac{dn}{dx}$$

$$J_p = -qD_p \frac{dp}{dx}$$



Notation:

$D_p \equiv$ hole diffusion constant (cm²/s)

$D_n \equiv$ electron diffusion constant (cm²/s)

Current Flow: Diffusion



- Diffusion current within a semiconductor consists of hole and electron components:

$$J_{p,diff} = -qD_p \frac{dp}{dx} \quad J_{n,diff} = qD_n \frac{dn}{dx}$$

$$J_{tot,diff} = q(D_n \frac{dn}{dx} - D_p \frac{dp}{dx})$$

Its total current density consists of both electrons and holes

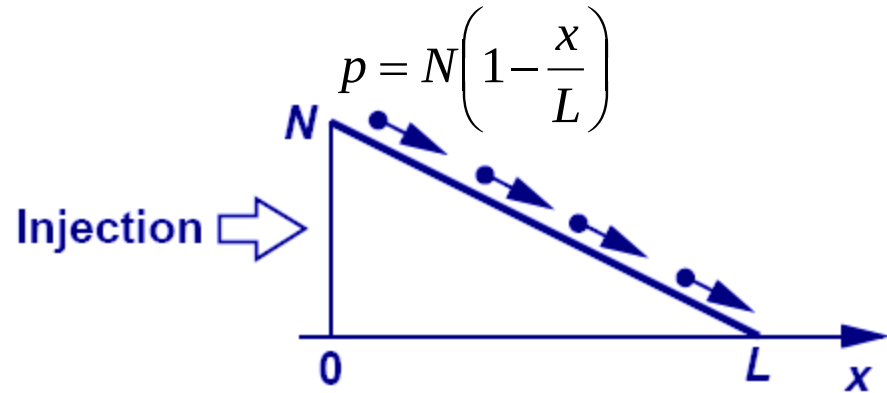
- The total current flowing in a semiconductor is the sum of drift current and diffusion current:

$$J_{tot} = J_{p,drift} + J_{n,drift} + J_{p,diff} + J_{n,diff}$$

Diffusion Examples

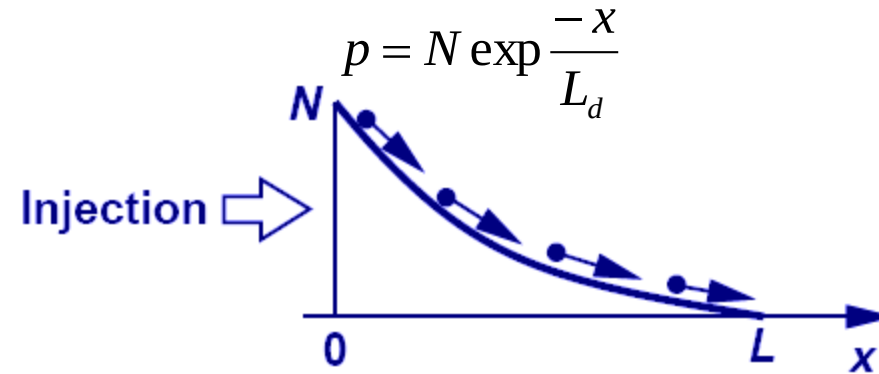


- Linear concentration profile
→ constant diffusion current



$$J_{p,diff} = -qD_p \frac{dp}{dx}$$
$$= qD_p \frac{N}{L}$$

- Non-linear concentration profile
→ varying diffusion current

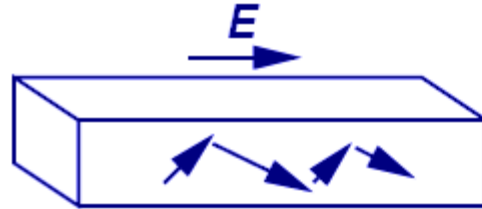


$$J_{p,diff} = -qD_p \frac{dp}{dx}$$
$$= \frac{qD_p N}{L_d} \exp \frac{-x}{L_d}$$

The Einstein Relation



Drift Current



$$J_n = q \mu_n E$$

$$J_p = q \mu_p E$$

Diffusion Current



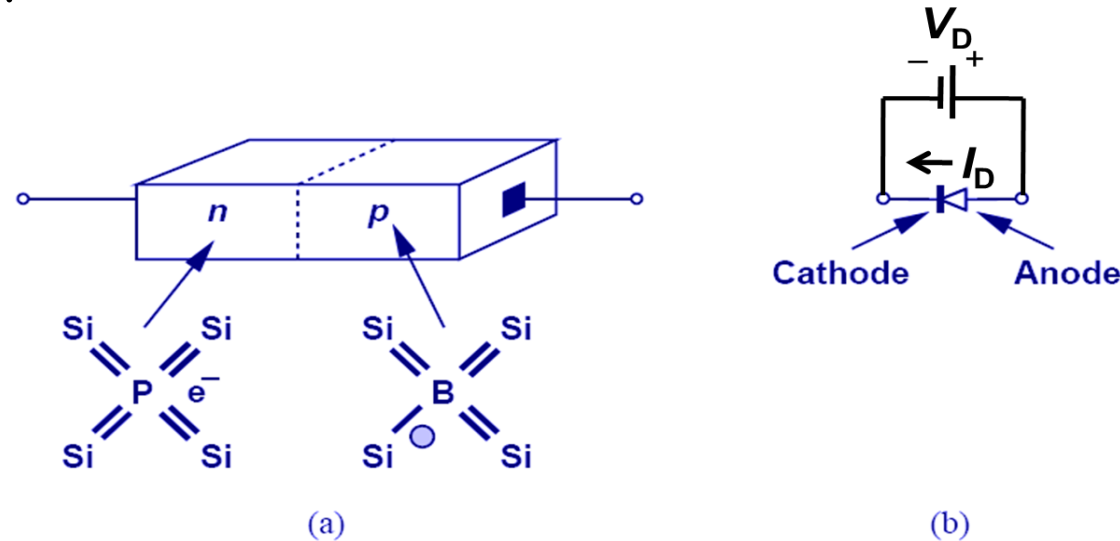
$$J_n = q D_n \frac{dn}{dx}$$

$$J_p = -q D_p \frac{dp}{dx}$$

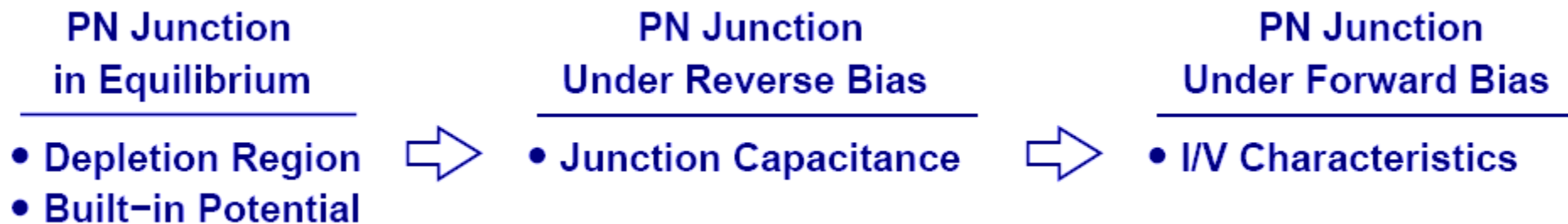
$$\frac{D}{\mu} = \frac{kT}{q}$$

- While the underlying physics behind drift and diffusion currents are totally different, Einstein's relation provides a mysterious link between the two.

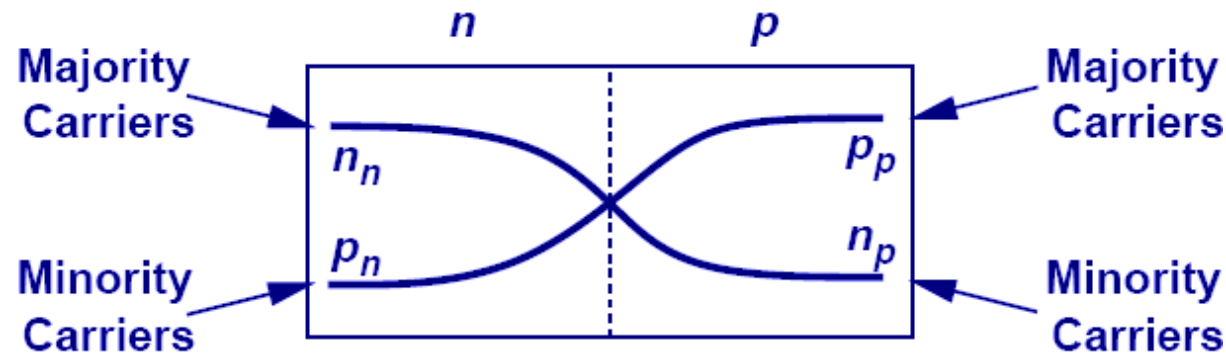
- When a P-type semiconductor region and an N-type semiconductor region are in contact, a PN junction diode is formed.



- In order to understand the operation of a diode, it is necessary to study its behavior in three operation regions: equilibrium, reverse bias, and forward bias.



Current Flow Across Junction: Diffusion



n_n : Concentration of electrons
on n side

p_n : Concentration of holes
on n side

p_p : Concentration of holes
on p side

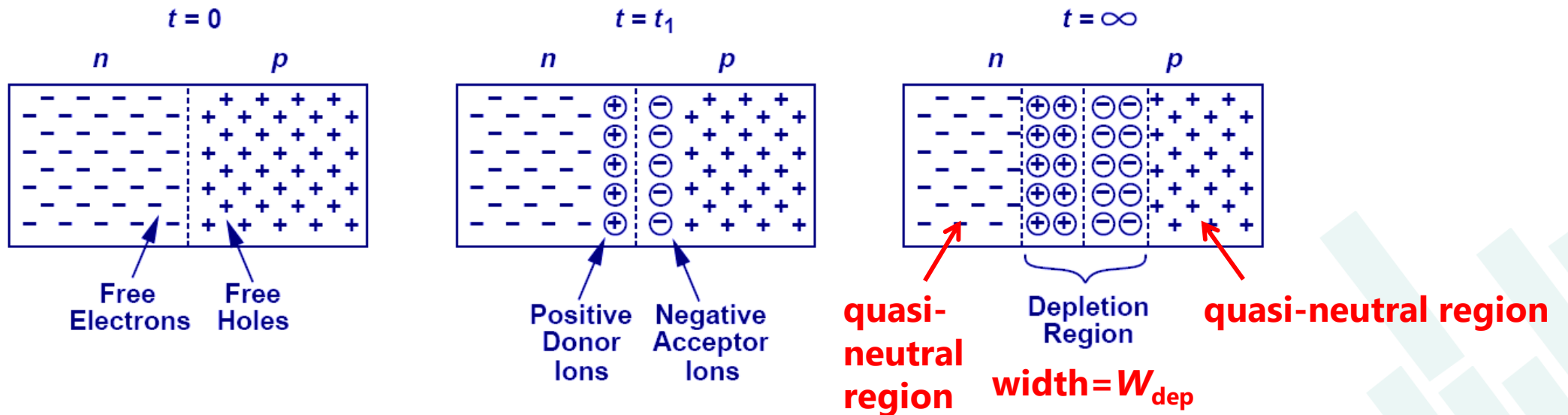
n_p : Concentration of electrons
on p side

- Because each side of the junction contains an excess of holes or electrons compared to the other side, there exists a large concentration gradient. Therefore, a diffusion current flows across the junction from each side.

Depletion Region



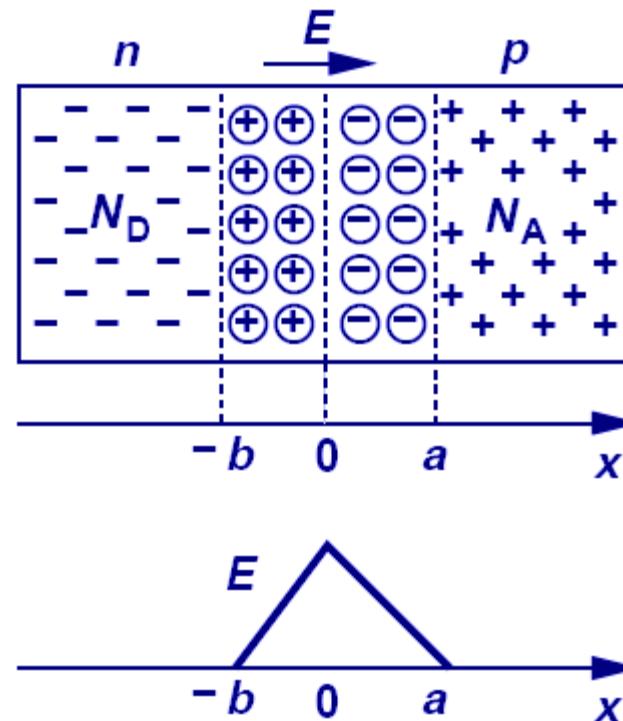
- As conduction electrons and holes diffuse across the junction, they leave behind ionized dopants. Thus, a region that is depleted of mobile carriers is formed.
 - The charge density in the depletion region is not zero.
 - The carriers which diffuse across the junction recombine with majority carriers, *i.e.* they are annihilated.



Carrier Drift across the Junction

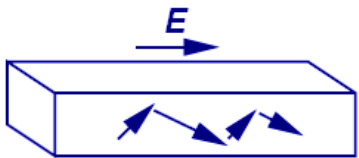


- Because charge density $\neq 0$ in the depletion region, an electric field exists, hence there is drift current.



pn diode energy band diagram

Drift Current



$$J_n = q \mu_n E$$

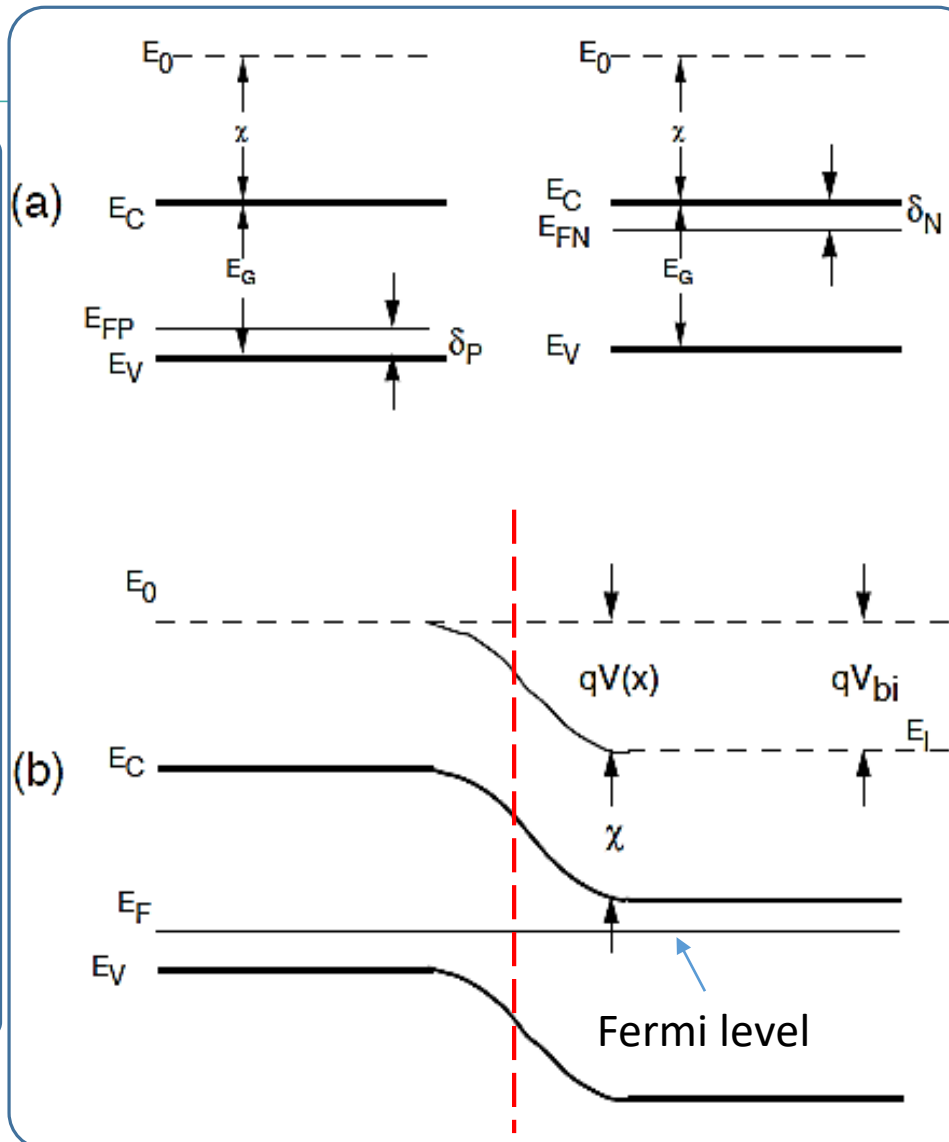
$$J_p = q \mu_p E$$

Diffusion Current

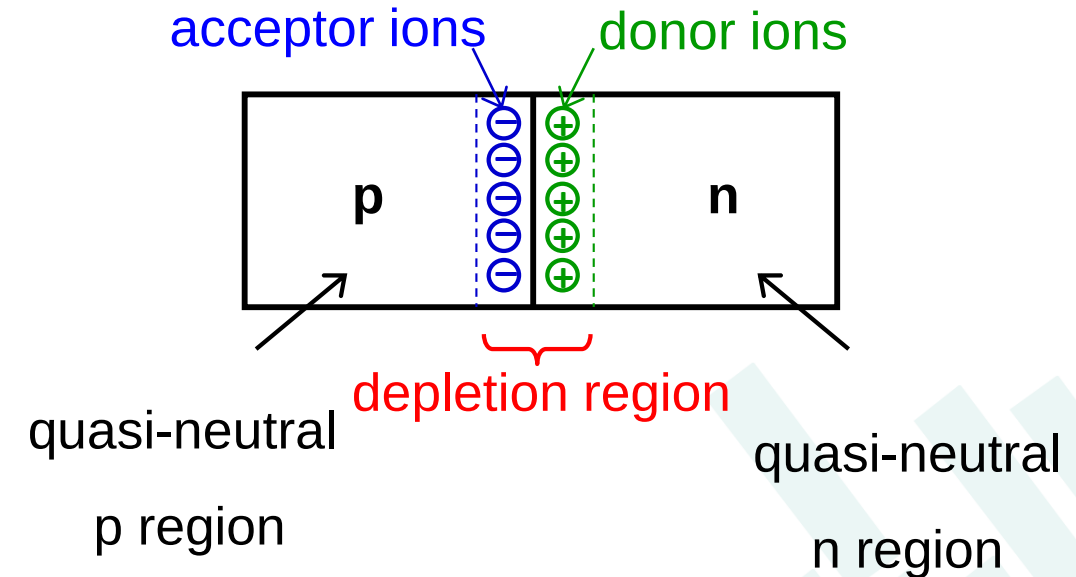


$$J_n = q D_n \frac{dn}{dx}$$

$$J_p = -q D_p \frac{dp}{dx}$$



- When the sides are “joined,” charges flow from one side to the other
- As electrons flow, they leave behind (positively charged) ionized donors
 - Donors cannot move
- Holes flow, leaving behind (negatively charged) ionized acceptors
- Separation of charges sets up electric field



In equilibrium, the drift and diffusion components of current are balanced; therefore the net current flowing across the junction is zero

PN Junction in Equilibrium



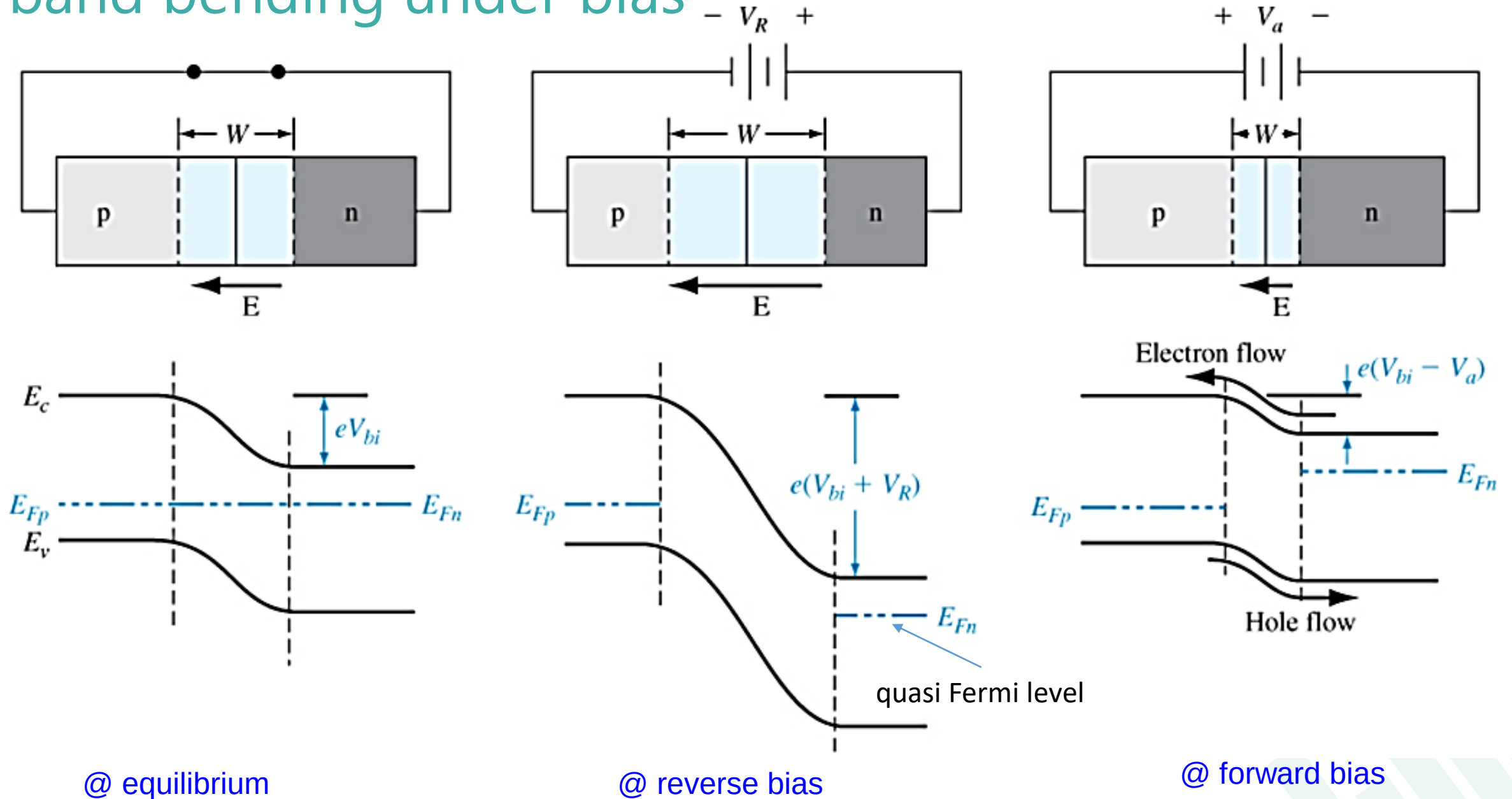
- In equilibrium, the drift and diffusion components of current are balanced; therefore the net current flowing across the junction is zero.

$$J_{p,drift} = -J_{p,diff}$$

$$J_{n,drift} = -J_{n,diff}$$

$$J_{tot} = J_{p,drift} + J_{n,drift} + J_{p,diff} + J_{n,diff} = 0$$

band bending under bias



Built-in Potential, V_0



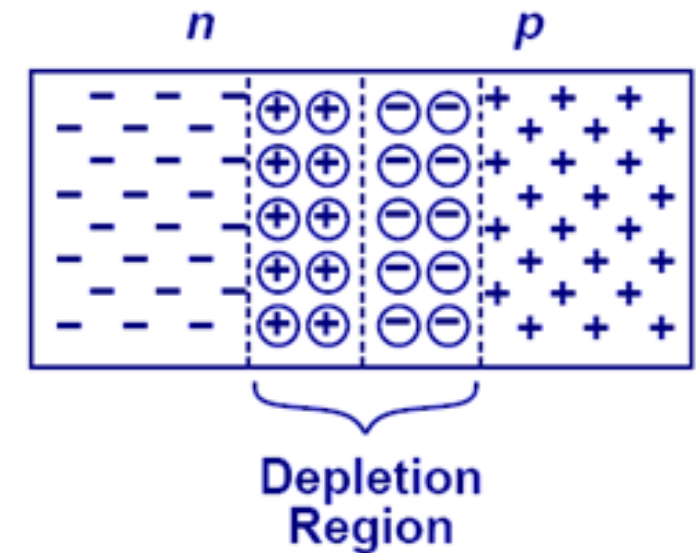
- Because of the electric field in the depletion region, there exists a potential drop across the junction:

$$qp\mu_p E = qD_p \frac{dp}{dx} \Rightarrow p\mu_p \left(-\frac{dV}{dx} \right) = D_p \frac{dp}{dx}$$

$$\Rightarrow -\mu_p \int_{x_1}^{x_2} dV = D_p \int_{p_n}^{p_p} \frac{dp}{p}$$

$$\Rightarrow V(x_1) - V(x_2) = \frac{D_p}{\mu_p} \ln \frac{p_p}{p_n} = \frac{kT}{q} \ln \left(\frac{N_A}{n_i^2 / N_D} \right)$$

$$V_{bi} = \frac{kT}{q} \ln \frac{N_A N_D}{n_i^2}$$



(Unit: Volts)

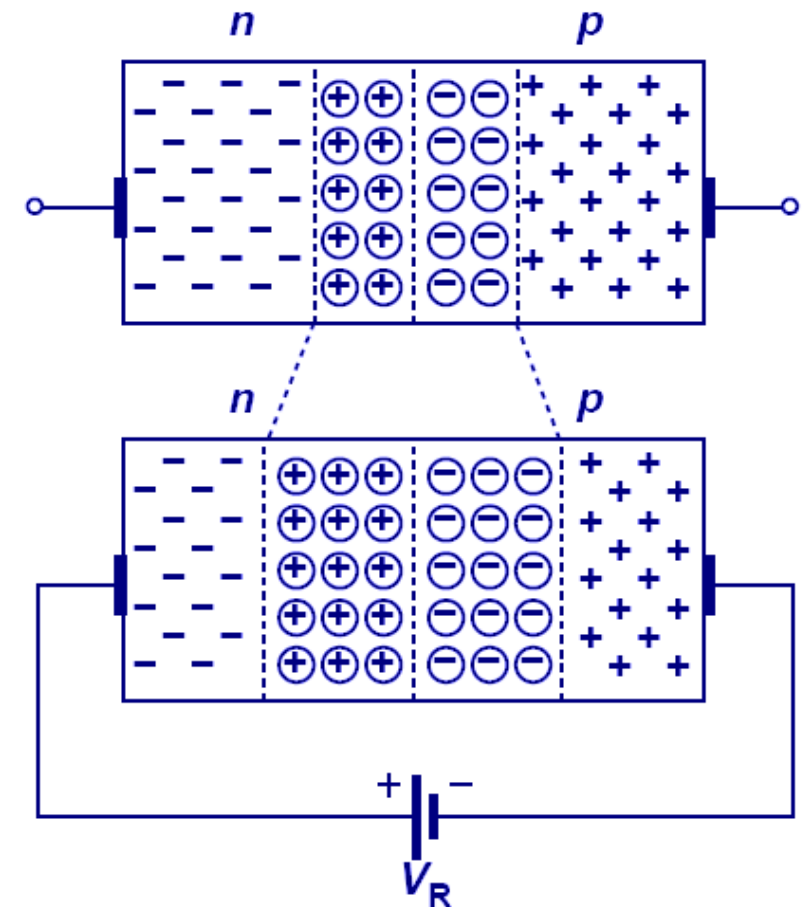
PN Junction under Reverse Bias



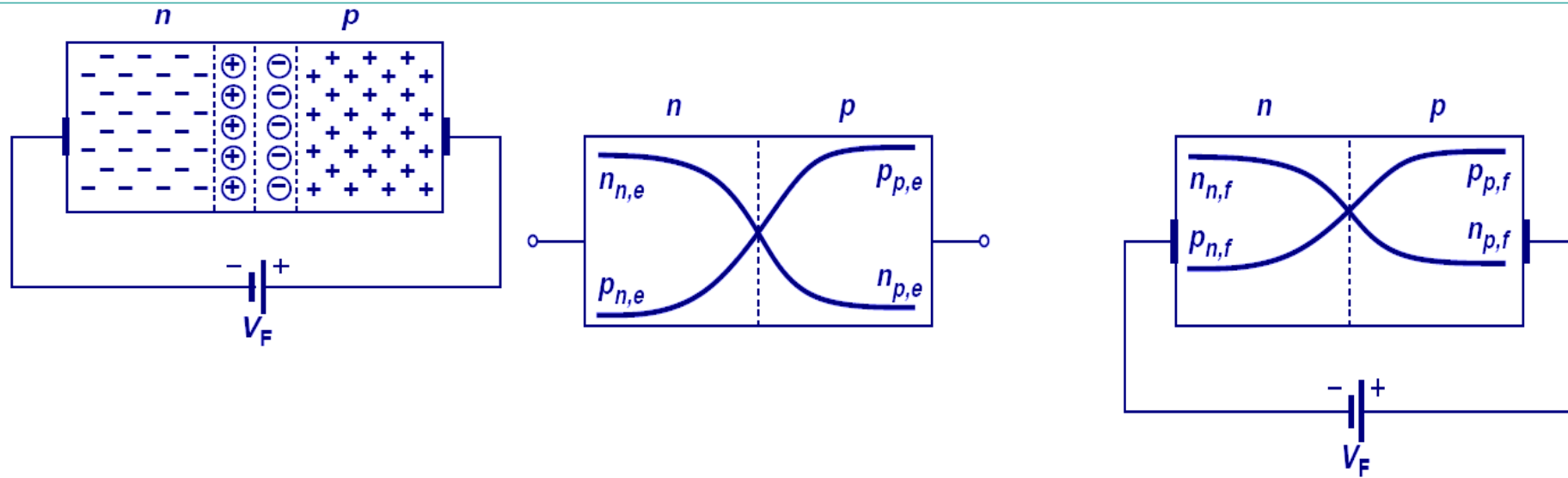
- A reverse bias increases the potential drop across the junction. As a result, the magnitude of the electric field increases and the width of the depletion region widens.

$$W_{dep} = \sqrt{\frac{2\epsilon_{si}}{q} \left(\frac{1}{N_A} + \frac{1}{N_D} \right) (V_0 + V_R)}$$

- In equilibrium, the **built-in potential effectively prevents carriers from diffusing across the junction.**
- Under reverse bias, the potential drop across the junction increases; therefore, negligible diffusion current flows. **A very small drift current flows**, limited by the rate at which minority carriers diffuse from the quasi-neutral regions into the depletion region.

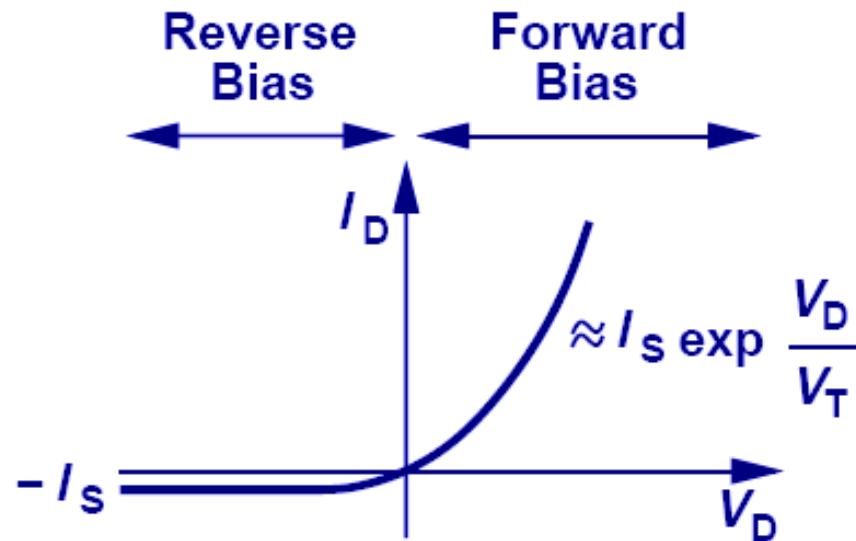


Diode in Forward Bias



- When the N-type region of a diode is at a lower potential than the P-type region, the diode is in forward bias.
- The **depletion width is shortened** and the **built-in electric field decreased**.
- Under forward bias, minority carriers in each region increase due to the **lowering of built-in field/potential**. Therefore, **diffusion currents increase** to supply these minority carriers.

IV Characteristic of PN Junction



$$I_D = I_s \left(\exp \frac{V_D}{V_T} - 1 \right)$$

$$I_s = Aq n_i^2 \left(\frac{D_n}{N_A L_n} + \frac{D_p}{N_D L_p} \right)$$

- The current and voltage relationship of a PN junction is exponential in forward bias region, and relatively constant in reverse bias region.